

таблицы

October 2026

1 Introduction

Таблица 1: CIFAR-10N: test accuracy (%). Модель ResNet-34.

Method / Noise	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
<i>Softmax</i>						
clean	95.34	95.53	95.34	95.41	95.58	95.40 $\pm$ 0.11
random1	87.42	87.99	87.44	87.96	88.70	87.90 $\pm$ 0.52
random2	88.06	88.75	88.46	88.14	88.14	88.31 $\pm$ 0.29
random3	86.56	87.91	87.25	88.65	87.93	87.66 $\pm$ 0.79
worst	80.12	81.06	81.17	80.28	77.66	80.06 $\pm$ 1.42
<i>Q-Softmax</i>						
clean	95.00 (1.05)	95.52 (1.11)	95.47 (1.26)	95.52 (1.01)	95.28 (1.34)	95.36 $\pm$ 0.22
random1	90.71 (1.84)	90.49 (1.82)	90.63 (1.82)	90.85 (1.87)	90.45 (1.85)	90.63 $\pm$ 0.16
random2	89.89 (1.87)	91.21 (1.94)	90.81 (1.86)	91.13 (1.96)	90.43 (1.76)	90.69 $\pm$ 0.55
random3	90.32 (1.91)	90.24 (1.92)	90.65 (1.96)	90.60 (1.86)	89.77 (1.71)	90.32 $\pm$ 0.35
worst	79.45 (0.93)	81.27 (1.95)	81.87 (1.73)	81.60 (1.78)	79.33 (1.2)	80.70 $\pm$ 1.22

Таблица 2: CIFAR-100N: test accuracy (%). Модель ResNet-34.

Method / Noise	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
<i>Softmax</i>						
clean	77.87	76.84	77.18	77.63	77.59	77.42 $\pm$ 0.41
noisy	54.87	54.80	54.80	54.72	54.31	54.70 $\pm$ 0.22
<i>Q-Softmax</i>						
clean	77.71 (0.98)	77.64 (1.41)	77.25 (1.39)	77.99 (1.14)	77.60 (0.99)	77.64 $\pm$ 0.27
noisy	62.07 (1.75)	62.32 (1.99)	61.98 (1.86)	62.22 (1.66)	61.93 (1.82)	62.10 $\pm$ 0.16

Таблица 3: Synthetic uniform noise: test accuracy (%). Модель ResNet-34.

Method	Noise η	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
<i>Softmax</i> — <i>CIFAR-10</i>							
	0.20	89.68	90.23	89.64	90.01	89.70	89.85 $\pm$ 0.26
	0.40	86.40	85.94	85.06	86.59	85.65	85.92 $\pm$ 0.51
	0.60	78.01	78.72	78.81	79.34	78.96	78.77 $\pm$ 0.49
	0.80	56.05	54.73	54.73	56.05	54.38	58.19 $\pm$ 0.80
<i>Q-Softmax</i> — <i>CIFAR-10</i>							
	0.20	91.68 (1.67)	93.04 (1.92)	92.29 (1.87)	92.74 (1.76)	93.03 (1.92)	92.56 $\pm$ 0.58
	0.40	86.32 (1.64)	87.38 (1.79)	88.06 (1.95)	88.54 (1.84)	87.79 (1.92)	87.62 $\pm$ 0.84
	0.60	80.32 (1.44)	80.45 (1.51)	80.35 (1.79)	80.18 (0.85)	80.75 (1.68)	80.41 $\pm$ 0.21
	0.80	57.57 (1.72)	54.19 (1.22)	55.27 (1.79)	56.99 (1.92)	53.34 (1.40)	55.47 $\pm$ 1.80
<i>Softmax</i> — <i>CIFAR-100</i>							
	0.20	63.70	64.28	64.57	64.09	64.60	64.25 $\pm$ 0.37
	0.40	55.04	53.90	53.49	52.75	53.20	53.68 $\pm$ 0.87
	0.60	45.12	46.02	46.34	46.04	43.35	45.37 $\pm$ 1.22
	0.80	23.11	22.17	23.36	24.25	22.80	23.14 $\pm$ 0.77
<i>Q-Softmax</i> — <i>CIFAR-100</i>							
	0.20	73.49 (1.73)	72.49 (1.78)	73.84 (1.70)	73.16 (1.77)	73.15 (1.65)	73.23 $\pm$ 0.50
	0.40	66.25 (1.60)	68.26 (1.75)	68.87 (1.73)	69.26 (1.76)	67.56 (1.66)	68.04 $\pm$ 1.19
	0.60	60.70 (1.92)	61.00 (1.91)	59.11 (1.76)	61.34 (1.96)	60.66 (1.94)	60.56 $\pm$ 0.86
	0.80	32.58 (1.98)	33.33 (1.98)	33.41 (1.96)	33.25 (1.84)	32.05 (1.83)	32.92 $\pm$ 0.59

Таблица 4: CIFAR-10N, $q = 1.8$: test accuracy (%). Модель ResNet-34.

Noise	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
clean	94.48	95.11	94.78	94.84	94.16	94.67 $\pm$ 0.36
random1	90.69	90.18	90.21	90.47	90.03	90.32 $\pm$ 0.26
random2	90.15	90.80	90.80	90.69	90.16	90.52 $\pm$ 0.34
random3	89.83	90.50	90.34	90.33	90.05	90.21 $\pm$ 0.27
worst	81.59	81.94	81.84	82.28	82.41	82.01 $\pm$ 0.33

Таблица 5: CIFAR-100N, $q = 1.8$: test accuracy (%). Модель ResNet-34.

Noise	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
clean	74.88	75.54	74.88	75.26	75.30	75.17 $\pm$ 0.26
noisy	62.69	62.25	62.30	62.86	61.88	62.40 $\pm$ 0.35

Таблица 6: Synthetic uniform noise, CIFAR-10, $q = 1.8$: test accuracy (%).

Noise η	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
0.20	92.10	92.72	92.19	92.44	91.99	92.29 $\pm$ 0.29
0.40	88.03	86.97	87.50	88.53	86.69	87.54 $\pm$ 0.75
0.60	80.88	81.08	80.35	79.91	81.06	80.66 $\pm$ 0.51
0.80	54.65	54.50	53.72	55.40	46.95	53.04 $\pm$ 3.46

Таблица 7: Synthetic uniform noise, CIFAR-100, $q = 1.8$: test accuracy (%).

Noise η	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
0.20	72.83	72.92	73.18	72.98	72.83	72.95 $\pm$ 0.14
0.40	68.91	69.06	69.20	68.22	68.49	68.78 $\pm$ 0.41
0.60	59.14	59.96	59.72	58.39	59.01	59.24 $\pm$ 0.62
0.80	32.20	31.61	30.81	33.04	31.50	31.83 $\pm$ 0.84

Таблица 8: CIFAR-10N, $q = 2.0$: test accuracy (%). Модель ResNet-34.

Noise	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
clean	-	-	-	-	-	
random1	90.98	90.83	91.05	91.30	90.98	91.02 $\pm$ 0.17
random2	90.06	91.25	91.22	90.72	91.42	90.93 $\pm$ 0.55
random3	90.95	90.80	90.87	90.81	90.93	90.87 $\pm$ 0.07
worst	81.35	83.29	82.16	81.20	83.17	82.23 $\pm$ 0.98

Таблица 9: CIFAR-100N, $q = 2.0$: test accuracy (%). Модель ResNet-34.

Noise	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
clean	-	-	-	-	-	
noisy	62.33	61.88	62.58	62.21	61.44	62.09 $\pm$ 0.44

Таблица 10: Synthetic uniform noise, CIFAR-10, $q = 2.0$: test accuracy (%).

Noise η	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
0.20	92.80	92.39	92.34	92.45	92.63	92.25 $\pm$ 0.19
0.40	88.64	88.42	88.37	89.15	88.18	88.55 $\pm$ 0.37
0.60	80.42	80.66	80.55	81.85	81.38	80.97 $\pm$ 0.62
0.80	56.66	56.08	55.73	57.25	53.73	55.89 $\pm$ 1.34

Таблица 11: Synthetic uniform noise, CIFAR-100, $q = 2.0$: test accuracy (%).

Noise η	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
0.20	-	-	-	-	-	
0.40	68.46	68.48	68.01	68.07	67.63	68.13 $\pm$ 0.35
0.60	61.28	61.03	61.36	61.51	59.78	60.99 $\pm$ 0.70
0.80	33.88	33.69	32.16	34.50	33.80	33.61 $\pm$ 0.87

Таблица 12: CIFAR-10N, $q = 2.5$: test accuracy (%). Модель ResNet-34.

Noise	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
clean	-	-	-	-	-	
random1	90.99	90.91	91.16	91.28	91.08	91.08 $\pm$ 0.14
random2	90.59	91.25	91.08	91.15	90.94	91.00 $\pm$ 0.26
random3	90.89	90.96	91.10	91.22	91.11	91.06 $\pm$ 0.13
worst	84.74	84.18	84.36	84.14	84.26	84.34 $\pm$ 0.24

Таблица 13: CIFAR-100N, $q = 2.5$: test accuracy (%). Модель ResNet-34.

Noise	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
clean	-	-	-	-	-	
noisy	60.01	59.75	59.97	60.04	59.56	59.87 $\pm$ 0.21

Таблица 14: Synthetic uniform noise, CIFAR-10, $q = 2.5$: test accuracy (%).

Noise η	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
0.20	92.18	91.83	91.60	91.99	91.74	91.87 $\pm$ 0.22
0.40	89.40	89.45	89.52	89.88	89.45	89.54 $\pm$ 0.19
0.60	83.26	83.60	82.92	84.07	83.39	83.45 $\pm$ 0.43
0.80	58.02	55.35	55.74	58.33	53.55	56.20 $\pm$ 1.99

Таблица 15: Synthetic uniform noise, CIFAR-100, $q = 2.5$: test accuracy (%).

Noise η	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
0.20	68.61	68.86	68.93	68.60	68.84	68.77 $\pm$ 0.15
0.40	65.99	66.26	66.79	65.76	64.43	65.85 $\pm$ 0.88
0.60	59.03	60.35	60.28	59.97	59.19	59.76 $\pm$ 0.62
0.80	36.05	35.32	33.85	35.94	35.77	35.39 $\pm$ 0.90

Таблица 16: CIFAR-10N, $q = 3.0$: test accuracy (%). Модель ResNet-34.

Noise	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
clean	-	-	-	-	-	-
random1	89.47	89.55	89.86	89.65	89.74	89.65 $\pm$ 0.15
random2	89.55	89.80	89.53	89.43	89.74	89.61 $\pm$ 0.15
random3	90.27	89.66	89.53	89.50	89.81	89.75 $\pm$ 0.31
worst	84.43	84.95	84.86	84.26	84.75	84.65 $\pm$ 0.29

Таблица 17: CIFAR-100N, $q = 3.0$: test accuracy (%). Модель ResNet-34.

Noise	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
clean	-	-	-	-	-	-
noisy	57.68	57.84	58.41	58.27	57.86	58.01 $\pm$ 0.31

Таблица 18: Synthetic uniform noise, CIFAR-10, $q = 3.0$: test accuracy (%).

Noise η	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
0.20	90.31	90.22	90.56	90.62	90.44	90.43 $\pm$ 0.17
0.40	88.50	88.41	88.91	88.87	88.21	88.58 $\pm$ 0.30
0.60	83.00	84.33	83.63	83.92	84.27	83.83 $\pm$ 0.54
0.80	57.58	54.48	57.11	58.21	50.30	55.54 $\pm$ 3.25

Таблица 19: Synthetic uniform noise, CIFAR-100, $q = 3.0$: test accuracy (%).

Noise η	seed 0	seed 1	seed 42	seed 123	seed 777	mean $\pm$ std
0.20	67.15	67.72	67.44	67.08	66.94	67.27 $\pm$ 0.31
0.40	63.95	65.02	64.28	64.52	63.85	64.32 $\pm$ 0.47
0.60	58.18	58.51	58.61	58.06	57.51	58.17 $\pm$ 0.44
0.80	35.88	33.76	31.61	34.04	35.68	34.19 $\pm$ 1.73

Таблица 20: CIFAR-10N: точность на тесте (%). Модель ResNet-34.

Шум	Optuna	$q = 1.8$	$q = 2.0$	$q = 2.5$	$q = 3.0$
clean	95.36 $\pm$ 0.22	94.67 $\pm$ 0.36	–	–	–
random1	90.63 $\pm$ 0.16	90.32 $\pm$ 0.26	91.02 $\pm$ 0.17	91.08 $\pm$ 0.14	89.65 $\pm$ 0.15
random2	90.69 $\pm$ 0.55	90.52 $\pm$ 0.34	90.93 $\pm$ 0.55	91.00 $\pm$ 0.26	89.61 $\pm$ 0.15
random3	90.32 $\pm$ 0.35	90.21 $\pm$ 0.27	90.87 $\pm$ 0.07	91.06 $\pm$ 0.13	89.75 $\pm$ 0.31
worst	80.70 $\pm$ 1.22	82.01 $\pm$ 0.33	82.23 $\pm$ 0.98	84.34 $\pm$ 0.24	84.65 $\pm$ 0.29

Таблица 21: CIFAR-100N: точность на тесте (%). Модель ResNet-34.

Шум	Optuna	$q = 1.8$	$q = 2.0$	$q = 2.5$	$q = 3.0$
clean	77.64 $\pm$ 0.27	75.17 $\pm$ 0.26	–	–	–
noisy	62.10 $\pm$ 0.16	62.40 $\pm$ 0.35	62.09 $\pm$ 0.44	59.87 $\pm$ 0.21	58.01 $\pm$ 0.31

Таблица 22: Синтетический равномерный шум, CIFAR-10: точность на тесте (%). Модель ResNet-34.

Шум η	Optuna	$q = 1.8$	$q = 2.0$	$q = 2.5$	$q = 3.0$
0.20	92.56 $\pm$ 0.58	92.29 $\pm$ 0.29	92.25 $\pm$ 0.19	91.87 $\pm$ 0.22	90.43 $\pm$ 0.17
0.40	87.62 $\pm$ 0.84	87.54 $\pm$ 0.75	88.55 $\pm$ 0.37	89.54 $\pm$ 0.19	88.58 $\pm$ 0.30
0.60	80.41 $\pm$ 0.21	80.66 $\pm$ 0.51	80.97 $\pm$ 0.62	83.45 $\pm$ 0.43	83.83 $\pm$ 0.54
0.80	55.47 $\pm$ 1.80	53.04 $\pm$ 3.46	55.89 $\pm$ 1.34	56.20 $\pm$ 1.99	55.54 $\pm$ 3.25

Таблица 23: Синтетический равномерный шум, CIFAR-100: точность на тесте (%). Модель ResNet-34.

Шум η	Optuna	$q = 1.8$	$q = 2.0$	$q = 2.5$	$q = 3.0$
0.20	73.23 $\pm$ 0.50	72.95 $\pm$ 0.14	–	68.77 $\pm$ 0.15	67.27 $\pm$ 0.31
0.40	68.04 $\pm$ 1.19	68.78 $\pm$ 0.41	68.13 $\pm$ 0.35	65.85 $\pm$ 0.88	64.32 $\pm$ 0.47
0.60	60.56 $\pm$ 0.86	59.24 $\pm$ 0.62	60.99 $\pm$ 0.70	59.76 $\pm$ 0.62	58.17 $\pm$ 0.44
0.80	32.92 $\pm$ 0.59	31.83 $\pm$ 0.84	33.61 $\pm$ 0.87	35.39 $\pm$ 0.90	34.19 $\pm$ 1.73

Таблица 24: CIFAR-10N (worst): точность на тесте (%). Модель ResNet-34.

Шум	$q = 2.5$	$q = 3.0$	$q = 3.5$
worst	80.03 $\pm$ 0.13	81.27 $\pm$ 0.18	80.99 $\pm$ 0.34

Таблица 25: CIFAR-100N (noisy): точность на тесте (%). Модель ResNet-34.

Шум	$q = 1.25$	$q = 1.5$	$q = 1.75$	$q = 2.0$	$q = 2.5$
noisy	52.06 $\pm$ 0.22	54.09 $\pm$ 0.49	55.18 $\pm$ 0.26	54.99 $\pm$ 0.45	53.53 $\pm$ 0.72

Таблица 26: Среднее время обучения на 80 эпох.

Метод	Время, с	Время, мин	На эпоху, с
Softmax	1965.2	32.75	24.57
Q-Softmax	1958.2	32.64	24.48
Разница (Q – CE)	-7.0	-0.11	-0.09

Таблица 27: Время обучения по seed'ам, секунды (80 эпох).

Seed	CE	Q-Softmax
42	1959	1914.6
123	1934	1923.6
777	1998	1988.4
0	2011	1990.8
1	1924	1973.4
Среднее	1965.2	1958.2